Chair of Network Architectures and Services
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Introductory Talk for the Bachelor's Thesis

Secure Delegation of zk-SNARK Proving to Cloud-based Trusted Execution Environments

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Begin:	2026-10-26
End:	2027-02-26

Motivation and Problem Statement

Zero-knowledge succinct non-interactive arguments of knowledge (zk-SNARKs) [1,2] allow a prover to convince a verifier that a public statement holds for some private witness without revealing the witness itself. Their properties make them useful across distinct application scenarios. In blockchain scaling, ZK-rollups use succinct proofs to verify batches of state transitions without requiring nodes to re-execute the complete computation [2,3]. In privacy-focused applications, zk-SNARKs enable confidential payments, where transaction validity can be proven without revealing sensitive transaction details, as well as digital identity systems, where users can prove properties such as age or membership without revealing the underlying identity data [1,2].

A major practical challenge is the asymmetry between verification and proof generation: while zk-SNARK proofs are succinct and can be verified efficiently, generating them can require substantial computational and memory resources [2,3]. On resource-constrained clients, local proof generation may therefore become impractical or even infeasible [2]. Delegating this computation to cloud infrastructure with sufficient resources is one possible approach to addressing this limitation [2].

However, naively outsourcing the computation introduces a confidentiality problem that undermines the privacy benefits of zk-SNARKs: the remote prover needs access to the plaintext witness in order to generate the proof [2]. The confidentiality of the witness therefore depends on the trustworthiness of the cloud provider and privileged host software, including the hypervisor, as well as on effective isolation from untrusted co-located virtual machines (VMs) sharing the same physical hardware [2,4,5]. Cloud-based delegation requires additional privacy-preserving measures whenever the witness must remain hidden from the underlying infrastructure and cloud provider.

To resolve this conflict, hardware-based isolation via trusted execution environments (TEEs) [4,5] can protect data in use from untrusted host infrastructure. This thesis investigates how cloud-based TEEs can enable secure delegation of zk-SNARK proof generation while keeping the private witness confidential. Beyond evaluating the security properties of this approach, the thesis measures the performance overhead of zk-SNARK proof generation within a TEE-based VM compared to a conventional VM.

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zk-SNARKs

The acronym *zk-SNARK* summarizes four defining properties of the proof system [1]:

- **Zero-Knowledge:** The proof does not reveal any information about the private witness beyond what can already be inferred from the public statement and its validity.
- **Succinct:** Proofs are compact and can be verified efficiently compared to the cost of evaluating the underlying computation.
- **Non-Interactive:** The prover generates a proof that can be verified without requiring an interactive exchange with the verifier for each proof.
- **Argument of Knowledge:** A successful prover demonstrates knowledge of a valid witness supporting the claimed statement, with soundness guaranteed against computationally bounded provers.

Trusted Execution Environments

Trusted execution environments provide hardware-backed isolation for sensitive code and data from software outside the protected environment [4, 5]. Their main security properties are *confidentiality* and *integrity*, which protect data and execution state, such as RAM contents and CPU register values, against unauthorized disclosure and modification, respectively [4, 5].

These properties are enforced through hardware mechanisms such as memory encryption, protected address translation, and page-ownership tracking [4, 5]. Modern confidential-computing technologies such as Intel TDX [5] and AMD SEV-SNP [4] extend these protections to entire VMs. A confidential VM can therefore protect sensitive workloads from software outside its trusted computing base (TCB), the set of hardware and software components on which its security depends [4, 5].

Remote attestation complements these protections by allowing a remote client to obtain cryptographic evidence about the measured state of the TEE and its execution environment and compare it against expected reference values [4, 5]. This verification mechanism provides a basis for the client to trust the TEE before sending sensitive data.

Proposed Approach

Figure 1 illustrates the proposed architecture for delegating zk-SNARK proof generation to a cloud-based TEE. The approach assumes that the TEE platform and the software running inside the confidential VM are trusted, while the surrounding cloud infrastructure and network are considered untrusted.

First, the client and the prover inside the TEE perform a Noise-based handshake [6] to establish an encrypted channel. Before transmitting any sensitive inputs, the client requests a fresh attestation report cryptographically bound to this channel. The client verifies the authenticity and freshness of the report, checks the environment measurements against trusted reference values, and validates the channel binding [4, 5]. Only if all checks succeed does the client transmit the private witness and public inputs to the TEE. Using a proving key prepared during a separate setup phase for the selected circuit [7], the prover generates the corresponding zk-SNARK proof and returns it to the client. The client can then submit the proof together with the public inputs to the verifier.

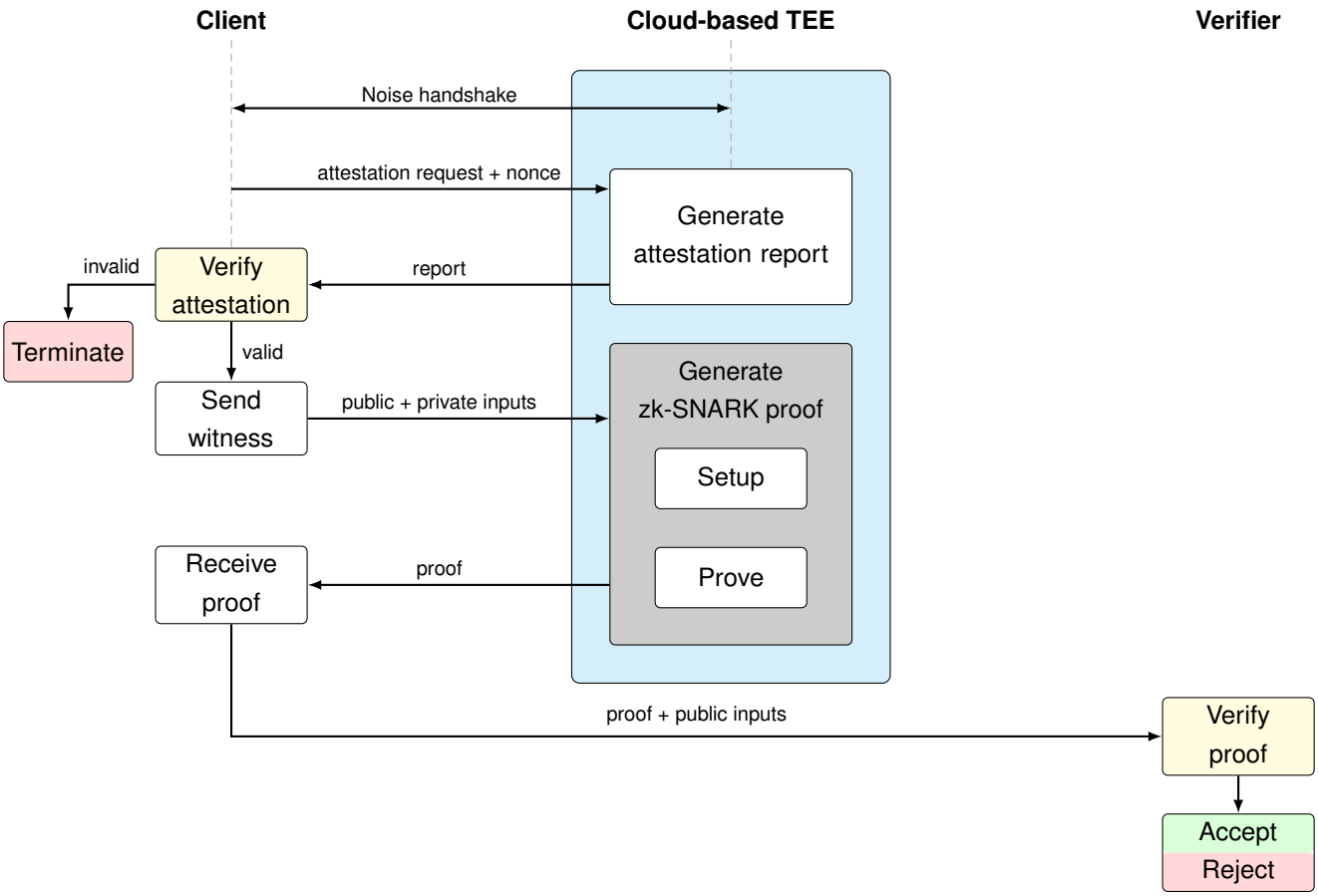


Figure 1: Workflow for secure delegation of zk-SNARK proving to a cloud-based TEE.

Research Questions

Based on the proposed architecture, this thesis addresses the following research questions:

RQ1: How can zk-SNARK proof generation be securely delegated to a cloud-based TEE while keeping the private witness confidential from the untrusted cloud infrastructure?

RQ2: How can remote attestation and a secure communication channel be combined to ensure that the private witness is transmitted only to the expected prover environment?

RQ3: What performance overhead does zk-SNARK proof generation inside a confidential VM introduce compared to a conventional, non-TEE VM?

Evaluation Plan

A security analysis will assess whether the proposed system protects the private witness both during transmission and during proof generation inside the TEE, taking into account the trust assumptions and potential limitations of the approach.

Performance will be evaluated by comparing zk-SNARK proof generation inside Intel TDX and AMD SEV-SNP confidential VMs against conventional VM baselines on the corresponding hardware platforms under comparable resource allocations. The benchmarks will use gnark [7], a Go library for zk-SNARKs, with the Groth16 and Plonk proof systems across multiple circuits of increasing computational complexity.

Proof generation time will serve as the primary performance metric. Memory consumption and CPU utilization will be measured to characterize the prover's and the client's resource usage. In addition, attestation latency and end-to-end latency will be evaluated to quantify the overhead of the complete delegation workflow.

The EnGINE framework [8] will be used to deploy the benchmark applications, automate repeated runs under controlled configurations, and collect measurement results. The configurations and software versions used will be recorded alongside the results to support reproducibility and comparability.

Aspect	**Evaluation**
Execution environment	Conventional VM vs. Intel TDX & AMD SEV-SNP
Proof systems	Groth16 and Plonk (gnark)
Workloads	Different circuit types and input sizes
Primary metric	Proof generation time
Additional metrics	Memory usage, CPU utilization, attestation latency, and end-to-end latency

Table 1: Planned performance evaluation of the proposed system.

Planned Schedule

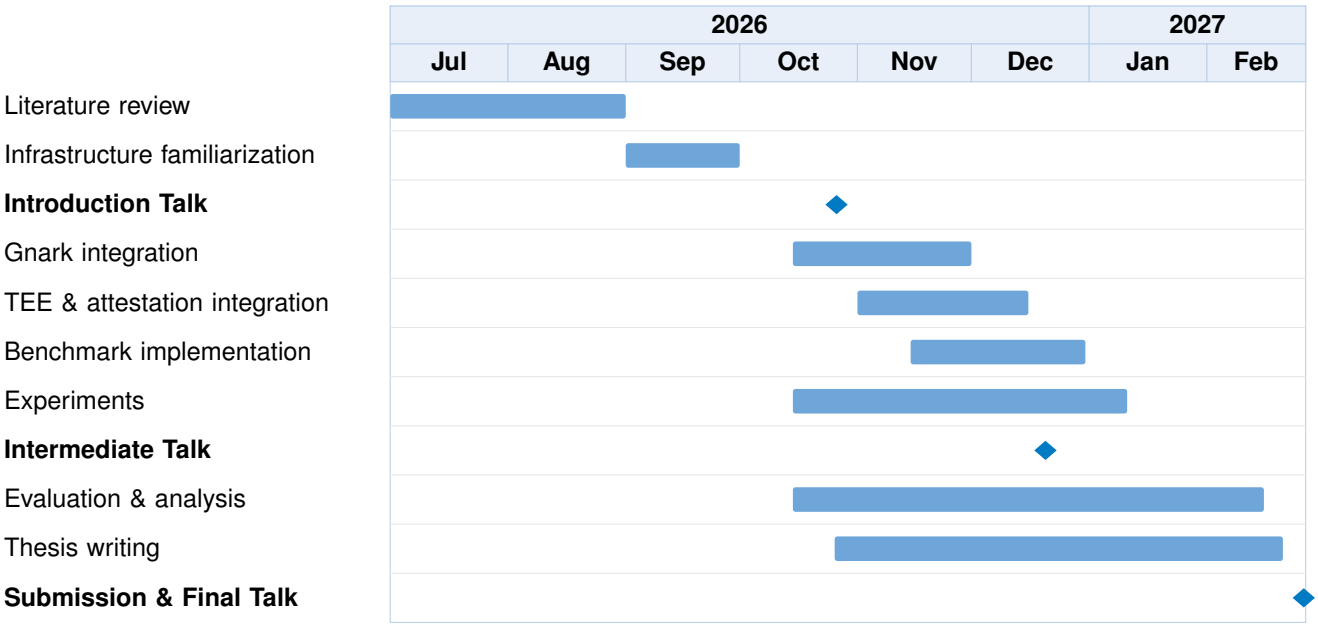


Figure 2: Planned schedule for the thesis.

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